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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 24th February Morning Shift

JEE Main Question Paper with Answer Key

24 Feb 2021	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	22 questions
Physics	27 questions
Out of syllabus	19 questions
Answer key	at the end

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Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

The electrode potential of M^{2+}/M of 3d-series elements shows positive value for :

(A) Zn (B) Fe (C) Cu (D) Co

Open solution

Q2. MCQ (Single Correct Answer)

Which of the following reagent is used for the following reaction ?

$$CH_3CH_2CH_3 \xrightarrow{?} CH_3CH_2CHO$$

(A) Molybdenum oxide (B) Manganese acetate

(C) Copper at high temperature and pressure (D) Potassium permanganate

Open solution

Q3. MCQ (Single Correct Answer)

Which of the following are isostructural pairs ?

A. SO_4^{2-} and CrO_4^{2-}

B. $SiCl_4$, and $TiCl_4$

C. NH_3 and NO_3^-

D. BCl_3 and $BrCl_3$

(A) B and C only (B) C and D only (C) A and B only (D) A and C only

Open solution

Q4. MCQ (Single Correct Answer)

Consider the elements Mg, Al, S, P and Si, the correct increasing order of their first ionization enthalpy is :

(A) Al < Mg < Si < S < P (B) Al < Mg < S < Si < P

(C) Mg < Al < Si < S < P (D) Mg < Al < Si < P < S

Open solution

Q5. MCQ (Single Correct Answer)

Out of the following, which type of interaction is responsible for the stabilisation of α-helix structure of proteins ?

(A) Covalent bonding (B) van der Waals forces

(C) Ionic bonding (D) Hydrogen bonding

Open solution

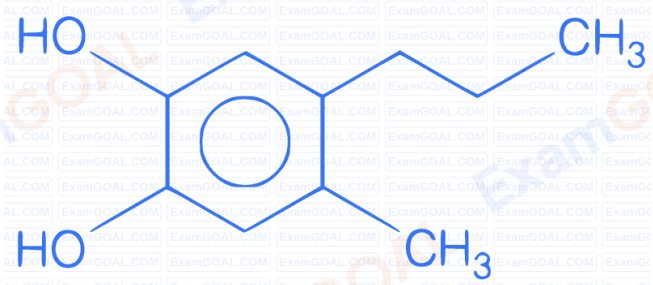
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Q6. MCQ (Single Correct Answer)

Open solution

Which of the following compound gives pink colour on reaction with phthalic anhydride in conc. H_2SO_4 followed by treatment with NaOH ?

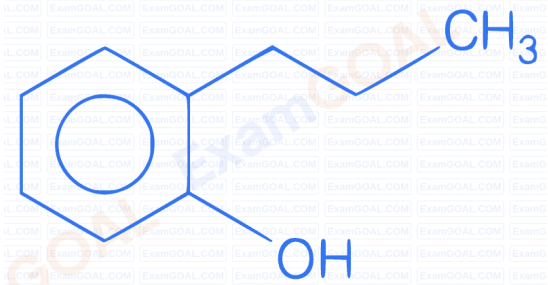
(A)



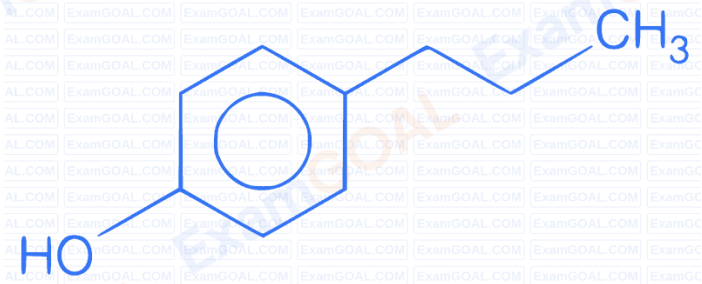
(B)



(C)



(D)



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Q7. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Colourless cupric metaborate is reduced to cuprous metaborate in a luminous flame.

Statement II : Cuprous metaborate is obtained by heating boric anhydride and copper sulphate in a non-luminous flame.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

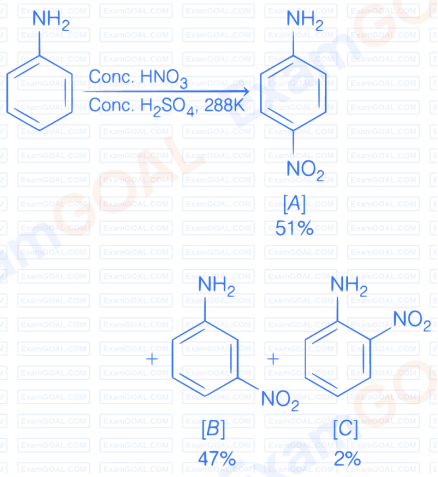
(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q8. MCQ (Single Correct Answer)

Open solution

In the following reaction the reason why meta-nitro product also formed is :



(A) $-NH_2$ group is highly meta-directive

(B) $-NO_2$ substitution always takes place at meta-position

(C) Formation of anilinium ion

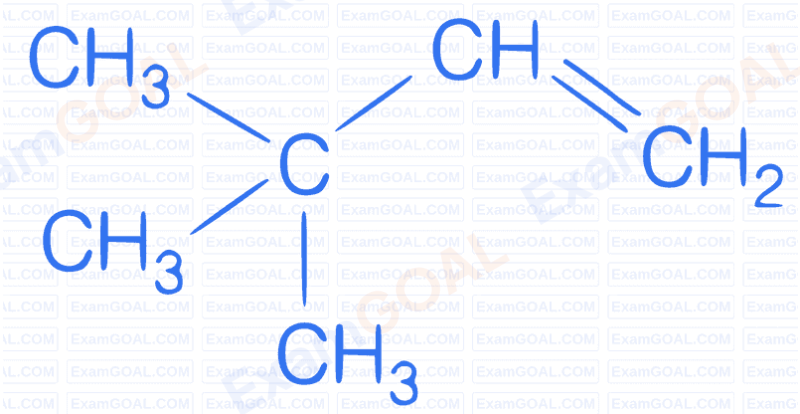
(D) low temperature

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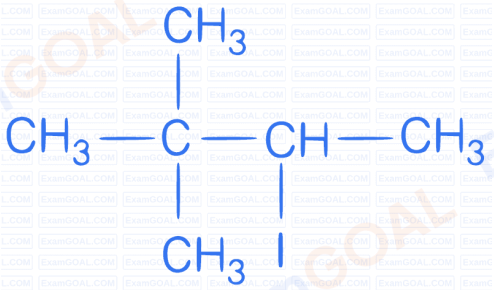
Q9. MCQ (Single Correct Answer)

Open solution

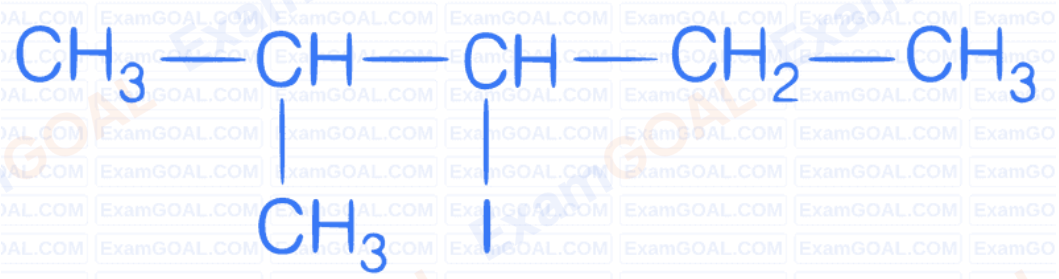
What is the major product formed by HI on reaction with



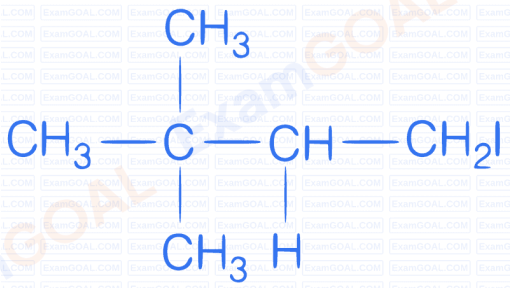
(A)



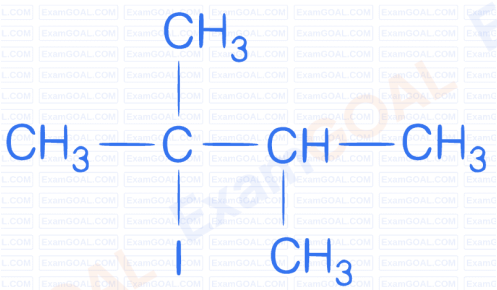
(B)



(C)



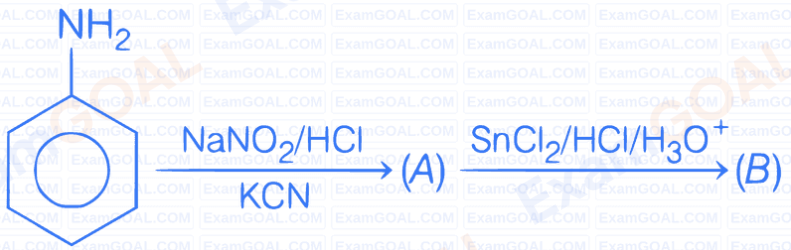
(D)



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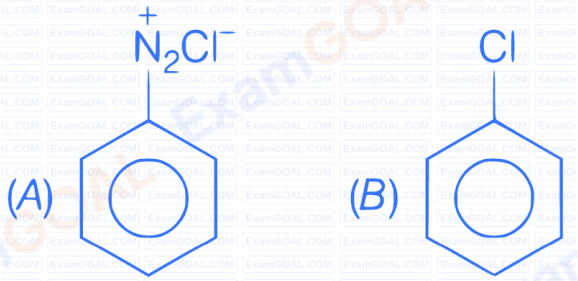
Q10. MCQ (Single Correct Answer)

'A' and 'B' in the following reactions are :

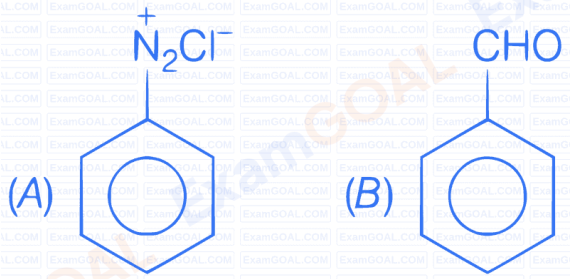


Open solution

(A)



(B)



(C)



(D)

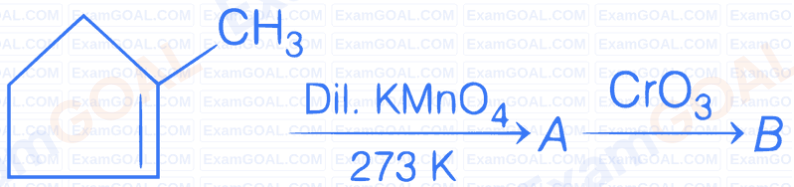


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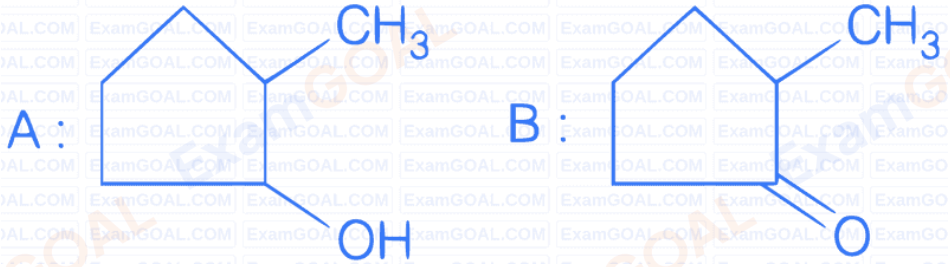
Q11. MCQ (Single Correct Answer)

Identify products A and B.

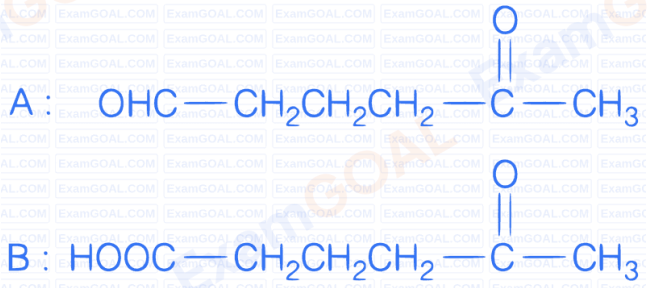
Open solution



(A)



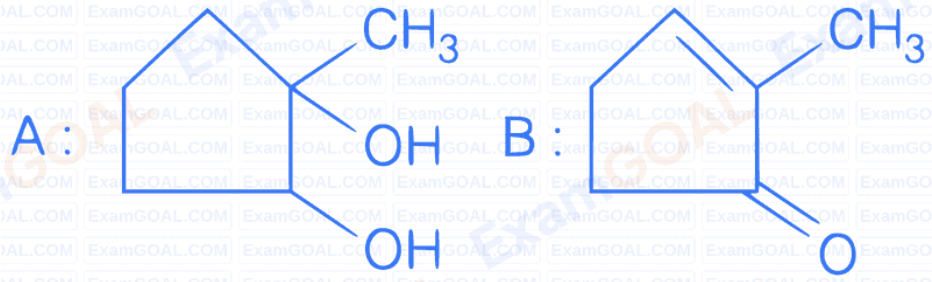
(B)



(C)



(D)

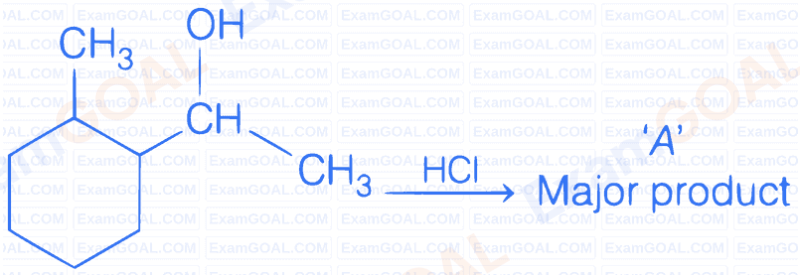


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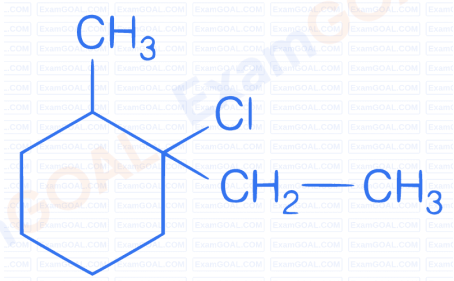
Q12. MCQ (Single Correct Answer)

Open solution

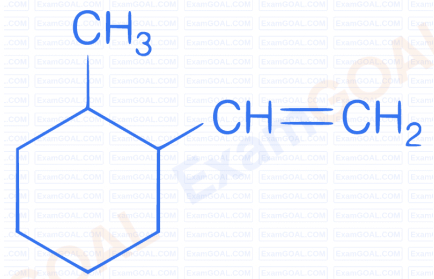
What is the final product (major) 'A' in the given reaction ?



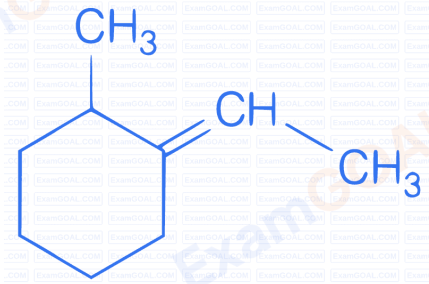
(A)



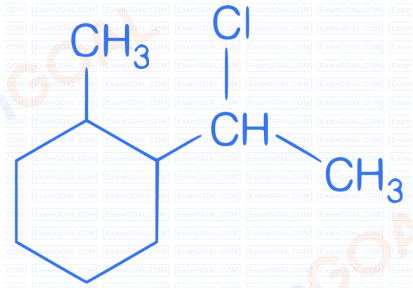
(B)



(C)



(D)

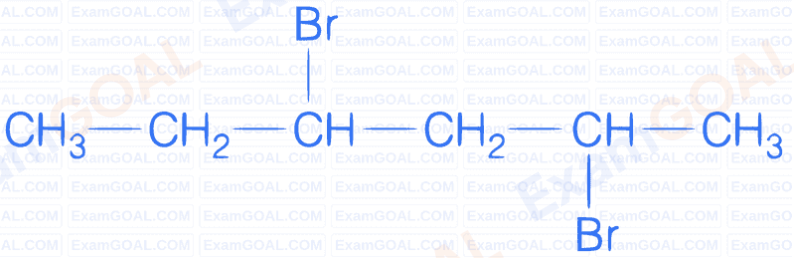


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Q13. MCQ (Single Correct Answer)

Open solution

The product formed in the first step of the reaction of

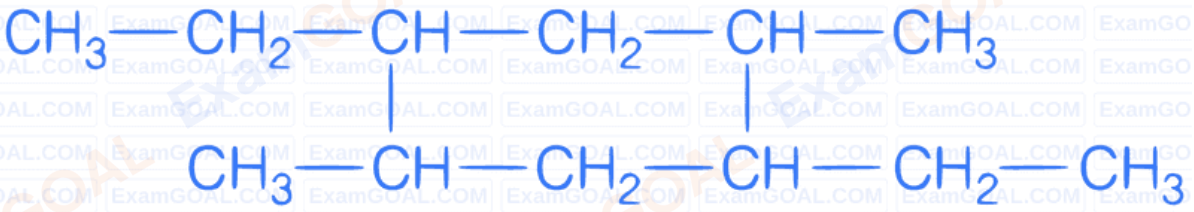


with excess Mg/Et_2O ($Et = C_2H_5$) is

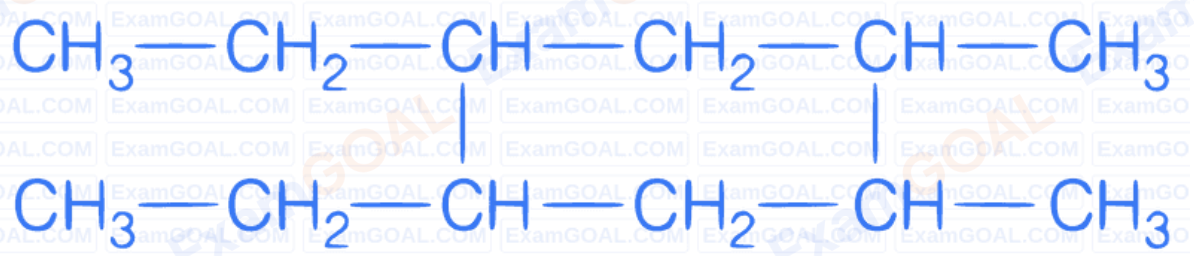
(A)



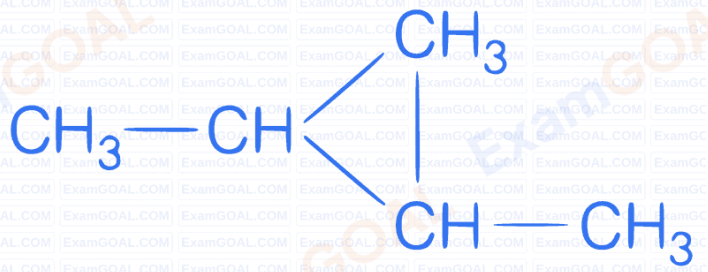
(B)



(C)



(D)



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Q14. Numerical

Number of amphoteric compounds among the following is _____.

(A) BeO

(B) BaO

(C) $Be(OH)_2$

(D) $Sr(OH)_2$

Open solution

Q15. Numerical

The stepwise formation of $[Cu(NH_3)_4]^{2+}$ is given below:

$$Cu^{2+} + NH_3 \overset{K_1}{\rightleftharpoons} [Cu(NH_3)]^{2+}$$

$$[Cu(NH_3)]^{2+} + NH_3 \overset{K_2}{\rightleftharpoons} [Cu(NH_3)_2]^{2+}$$

$$[Cu(NH_3)_2]^{2+} + NH_3 \overset{K_3}{\rightleftharpoons} [Cu(NH_3)_3]^{2+}$$

$$[Cu(NH_3)_3]^{2+} + NH_3 \overset{K_4}{\rightleftharpoons} [Cu(NH_3)_4]^{2+}$$

The value of stability constants K_1, K_2, K_3 and K_4 are 10^4, 1.58×10^3, 5×10^2 and 10^2 respectively.

The overall equilibrium constants for dissociation of $[Cu(NH_3)_4]^{2+}$ is $x \times 10^{-12}$.

The value of x is _____. (Rounded off to the nearest integer)

Open solution

Q16. Numerical

A proton and a Li^{3+} nucleus are accelerated by the same potential. If λ_{Li} and λ_p denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value of $\frac{\lambda_{Li}}{\lambda_p}$ is $x \times 10^{-1}$.

The value of x is _____. (Rounded off to the nearest integer)

[Mass of Li^{3+} = 8.3 mass of proton]

Open solution

Q17. Numerical

At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value of K_p for the reaction $Cl_{2\,(g)} \rightleftharpoons 2Cl_{(g)}$ under the above conditions is $x \times 10^{-1}$.

The value of x is _____. (Rounded off to the nearest integer)

Open solution

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Q18. Numerical

When 9.45 g of $ClCH_2COOH$ is added to 500 mL of water, its freezing point drops by 0.5°C. The dissociation constant of $ClCH_2COOH$ is $x \times 10^{-3}$.

The value of x is _____. (Rounded off to the nearest integer)

[$K_{f(H_2O)}$ = 1.86 K kg mol^{-1}]

Open solution

Q19. Numerical

4.5 g of compound A (MW = 90) was used to make 250 mL of its aqueous solution. The molarity of the solution in M is $x \times 10^{-1}$.

The value of x is _____. (Rounded off to the nearest integer)

Open solution

Q20. Numerical

The reaction of sulphur in alkaline medium is given below:

$$S_{8(s)} + a\ OH^-_{(aq)} \longrightarrow b\ S^{2-}_{(aq)} + c\ S_2O_3^{2-}_{(aq)} + d\ H_2O_{(l)}$$

The value of 'a' is _____. (Integer answer)

Open solution

Q21. Numerical

For the reaction $A_{(g)} \rightarrow B_{(g)}$ the value of the equilibrium constant at 300 K and 1 atm is equal to 100.0. The value of Δ_rG for the reaction at 300 K and 1 atm in J mol^{-1} is – xR, where x is _____. (Rounded off to the nearest integer)

[R = 8.31 J $mol^{-1}K^{-1}$ and ln 10 = 2.3]

Open solution

Q22. Numerical

Gaseous cyclobutene isomerises to butadiene in a first order process which has a 'k' value of 3.3×10^{-4} s^{-1} at 153°C. The time in minutes it takes for the isomerization to proceed 40% to completion at this temperature is _____.

(Rounded off to the nearest integer)

Open solution

Mathematics 22 questions

Q23. MCQ (Single Correct Answer)

If $\int \frac{\cos x - \sin x}{\sqrt{8 - \sin 2x}} dx = a\sin^{-1}\left(\frac{\sin x + \cos x}{b}\right) + c$, where c is a constant of integration, then the ordered pair (a, b) is equal to :

(A) (-1, 3) (B) (1, 3) (C) (1, -3) (D) (3, 1)

Open solution

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Q24. MCQ (Single Correct Answer)

If f : R $\rightarrow$ R is a function defined by f(x)= [x - 1] $\cos\left(\frac{2x-1}{2}\right)\pi$, where [.] denotes the greatest integer function, then f is :

(A) continuous for every real x

(B) discontinuous at all integral values of x except at x = 1

(C) discontinuous only at x = 1

(D) continuous only at x = 1

Open solution

Q25. MCQ (Single Correct Answer)

The population P = P(t) at time 't' of a certain species follows the differential equation

$\frac{dP}{dt}$ = 0.5P – 450. If P(0) = 850, then the time at which population becomes zero is :

(A) $\log_e 18$ (B) $\frac{1}{2}\log_e 18$ (C) $2\log_e 18$ (D) $\log_e 9$

Open solution

Q26. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be defined as f (x) = 2x – 1 and g : R - {1} $\rightarrow$ R be defined as g(x) = $\frac{x-\frac{1}{2}}{x-1}$. Then the composition function f(g(x)) is :

(A) one-one but not onto

(B) onto but not one-one

(C) both one-one and onto

(D) neither one-one nor onto

Open solution

Q27. MCQ (Single Correct Answer)

The function
f(x) = $\frac{4x^3-3x^2}{6} - 2\sin x + (2x-1)\cos x$:

(A) increases in $\left(-\infty, \frac{1}{2}\right]$

(B) decreases in $\left(-\infty, \frac{1}{2}\right]$

(C) increases in $\left[\frac{1}{2}, \infty\right)$

(D) decreases in $\left[\frac{1}{2}, \infty\right)$

Open solution

Q28. MCQ (Single Correct Answer)

$\lim\limits_{x\to 0} \frac{\int_0^{x^2}\left(\sin\sqrt{t}\right)dt}{x^3}$ is equal to :

(A) $\frac{1}{15}$ (B) 0 (C) $\frac{2}{3}$ (D) $\frac{3}{2}$

Open solution

Q29. MCQ (Single Correct Answer)

A scientific committee is to be formed from 6 Indians and 8 foreigners, which includes at least 2 Indians and double the number of foreigners as Indians. Then the number of ways, the committee can be formed, is :

(A) 1050 (B) 575 (C) 560 (D) 1625

Open solution

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Q30. MCQ (Single Correct Answer)

A man is walking on a straight line. The arithmetic mean of the reciprocals of the intercepts of this line on the coordinate axes is $\frac{1}{4}$. Three stones A, B and C are placed at the points (1, 1), (2, 2) and (4, 4) respectively. Then, which of these stones is / are on the path of the man?

(A) A only (B) All the three (C) C only (D) B only

Open solution

Q31. MCQ (Single Correct Answer)

Let p and q be two positive numbers such that p + q = 2 and p^4+q^4 = 272. Then p and q are roots of the equation :

(A) $x^2 - 2x + 8 = 0$ (B) $x^2 - 2x + 136=0$ (C) $x^2 - 2x + 16 = 0$ (D) $x^2 - 2x + 2 = 0$

Open solution

Q32. MCQ (Single Correct Answer)

The area (in sq. units) of the part of the circle $x^2 + y^2 = 36$, which is outside the parabola $y^2 = 9x$, is :

(A) $12\pi - 3\sqrt{3}$ (B) $24\pi + 3\sqrt{3}$ (C) $24\pi - 3\sqrt{3}$ (D) $12\pi + 3\sqrt{3}$

Open solution

Q33. MCQ (Single Correct Answer)

If $e^{(\cos^2 x+\cos^4 x+\cos^6 x+\ldots\infty)\log_e 2}$ satisfies the equation $t^2 - 9t + 8 = 0$, then the value of $\frac{2\sin x}{\sin x+\sqrt{3}\cos x}\left(0 < x < \frac{\pi}{2}\right)$ is :

(A) $\sqrt{3}$ (B) $\frac{3}{2}$ (C) $2\sqrt{3}$ (D) $\frac{1}{2}$

Open solution

Q34. MCQ (Single Correct Answer)

The locus of the mid-point of the line segment joining the focus of the parabola $y^2 = 4ax$ to a moving point of the parabola, is another parabola whose directrix is :

(A) x = 0 (B) x = $-\frac{a}{2}$ (C) x = a (D) x = $\frac{a}{2}$

Open solution

Q35. MCQ (Single Correct Answer)

The system of linear equations
3x - 2y - kz = 10
2x - 4y - 2z = 6
x+2y - z = 5m
is inconsistent if :

(A) $k \neq 3, m \in \mathbf{R}$ (B) $k = 3, m \neq \frac{4}{5}$
(C) $k = 3, m = \frac{4}{5}$ (D) $k \neq 3, m \neq \frac{4}{5}$

Open solution

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Q36. Numerical

Let P = $\begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in R$. Suppose Q = [q_{ij}] is a matrix satisfying PQ = kI_3 for some non-zero $k \in R$.

If $q_{23} = -\frac{k}{8}$ and |Q| = $\frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to _____.

Open solution

Q37. Numerical

The minimum value of α for which the equation $\frac{4}{\sin x} + \frac{1}{1-\sin x} = \alpha$ has at least one solution in $\left(0, \frac{\pi}{2}\right)$ is .....

Open solution

Q38. Numerical

If the least and the largest real values of α, for which the equation $z + \alpha|z - 1| + 2i = 0$ ($z \in C$ and $i = \sqrt{-1}$) has a solution, are p and q respectively; then $4(p^2 + q^2)$ is equal to _____.

Open solution

Q39. Numerical

$\lim\limits_{n\to\infty} \tan\left\{\sum\limits_{r=1}^{n} \tan^{-1}\left(\frac{1}{1+r+r^2}\right)\right\}$ is equal to _____.

Open solution

Q40. Numerical

Let A = {n ∈ N: n is a 3-digit number}

B = {9k + 2: k ∈ N}

and C = {9k + l: k ∈ N} for some $l(0 < l < 9)$

If the sum of all the elements of the set A ∩ (B ∪ C) is 274 × 400, then l is equal to _____.

Open solution

Q41. Numerical

If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C', whose center is at (2, 1), then its radius is _____.

Open solution

Q42. Numerical

If $\int\limits_{-a}^{a} (|x| + |x - 2|)dx = 22$, (a > 2) and [x] denotes the greatest integer $\leq$ x, then $\int\limits_{a}^{-a} (x + [x])dx$ is equal to _____.

Open solution

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Q43. Numerical

Let M be any 3 $\times$ 3 matrix with entries from the set {0, 1, 2}. The maximum number of such matrices, for which the sum of diagonal elements of M^TM is seven, is _____.

Open solution

Q44. Numerical

Let B_i (i = 1, 2, 3) be three independent events in a sample space. The probability that only B_1 occur is α, only B_2 occurs is β and only B_3 occurs is γ. Let p be the probability that none of the events B_i occurs and these 4 probabilities satisfy the equations $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$ (All the probabilities are assumed to lie in the interval (0, 1)).
Then $\frac{P(B_1)}{P(B_3)}$ is equal to _____.

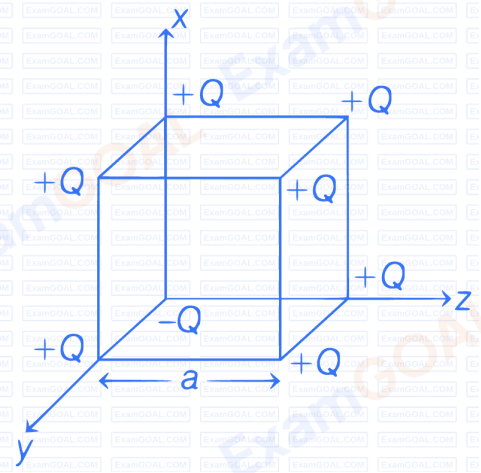
Open solution

Physics 27 questions

Q45. MCQ (Single Correct Answer)

A cube of side 'a' has point charges +Q located at each of its vertices except at the origin where the charge is $-$Q. The electric field at the centre of cube is :

Open solution



(A) $\frac{2Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

(B) $\frac{-Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

(C) $\frac{Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

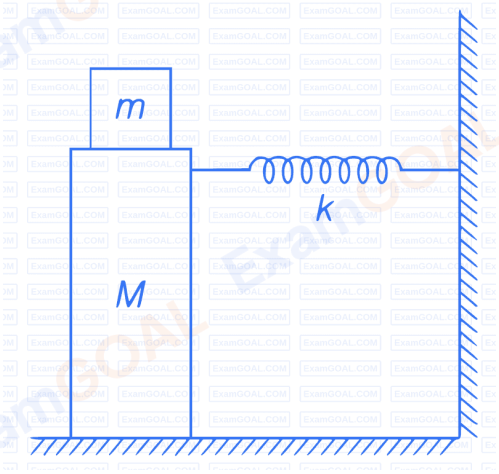
(D) $\frac{-2Q}{3\sqrt{3}\pi\varepsilon_0 a^2}\left(\hat{x}+\hat{y}+\hat{z}\right)$

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Q46. MCQ (Single Correct Answer)

Open solution

In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k. The mass oscillates on a frictionless surface with time period T and amplitude A. When the mass is in equilibrium position, as shown in the figure, another mass m is gently fixed upon it. The new amplitude of oscillations will be :



(A) $A\sqrt{\frac{M}{M-m}}$ (B) $A\sqrt{\frac{M-m}{M}}$ (C) $A\sqrt{\frac{M+m}{M}}$ (D) $A\sqrt{\frac{M}{M+m}}$

Q47. MCQ (Single Correct Answer)

Open solution

A current through a wire depends on time as

$i = \alpha_0 t + \beta t^2$

where α_0 = 20 A/s and β = 8 As^{-2}. Find the charge crossed through a section of the wire in 15 s.

(A) 2250 C (B) 2100 C (C) 260 C (D) 11250 C

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Q48. MCQ (Single Correct Answer)

Open solution

Match List I with List II.

	List I		List II
(a)	Isothermal	(i)	Pressure constant
(b)	Isochoric	(ii)	Temperature constant
(c)	Adiabatic	(iii)	Volume constant
(d)	Isobaric	(iv)	Heat content is constant

Choose the correct answer from the options given below :

(1) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(2) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(3) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

(4) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

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Q49. MCQ (Single Correct Answer)

Open solution

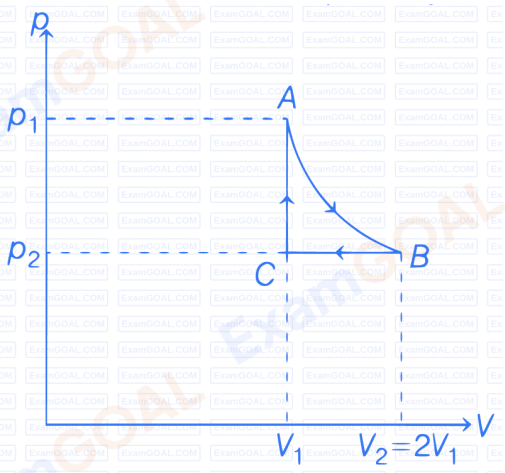
n mole of a perfect gas undergoes a cyclic process ABCA (see figure) consisting of the following processes.

A $\rightarrow$ B : Isothermal expansion at temperature T so that the volume is doubled from V_1 to $V_2 = 2V_1$ and pressure changes from P_1 to P_2

B $\rightarrow$ C : Isobaric compression at pressure P_2 to initial volume V_1.

C $\rightarrow$ A : Isochoric change leading to change of pressure from P_2 to P_1.

Total work done in the complete cycle ABCA is :



(A) nRTln 2

(B) 0

(C) $nRT\left(\ln 2 - \frac{1}{2}\right)$

(D) $nRT\left(\ln 2 + \frac{1}{2}\right)$

Q50. MCQ (Single Correct Answer)

Open solution

Each side of a box made of metal sheet in cubic shape is 'a' at room temperature 'T', the coefficient of linear expansion of the metal sheet is 'α'. The metal sheet is heated uniformly, by a small temperature ΔT, so that its new temperature is T + ΔT. Calculate the increase in the volume of the metal box.

(A) $3a^3\alpha\Delta T$

(B) $4\pi a^3\alpha\Delta T$

(C) $\frac{4}{3}\pi a^3\alpha\Delta T$

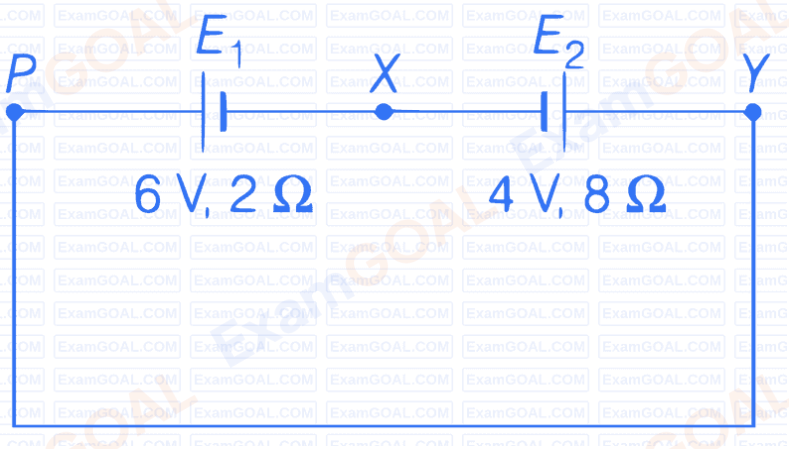
(D) $4a^3\alpha\Delta T$

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Q51. MCQ (Single Correct Answer)

A cell E_1 of emf 6V and internal resistance 2Ω is connected with another cell E_2 of emf 4V and internal resistance 8Ω (as shown in the figure). The potential difference across points X and Y is :

Open solution



(A) 10.0 V (B) 2.0 V (C) 5.6 V (D) 3.6 V

Q52. MCQ (Single Correct Answer)

The focal length f is related to the radius of curvature r of the spherical convex mirror by :

Open solution

(A) f = r (B) f = − r

(C) f = $+\frac{1}{2}$ r (D) f = $-\frac{1}{2}$ r

Q53. MCQ (Single Correct Answer)

In a Young's double slit experiment, the width of the one of the slit is three times the other slit. The amplitude of the light coming from a slit is proportional to the slit-width. Find the ratio of the maximum to the minimum intensity in the interference pattern.

Open solution

(A) 4 : 1 (B) 2 : 1 (C) 1 : 4 (D) 3 : 1

Q54. MCQ (Single Correct Answer)

Four identical particles of equal masses 1 kg made to move along the circumference of a circle of radius 1 m under the action of their own mutual gravitational attraction. The speed of each particle will be :

Open solution

(A) $\sqrt{\frac{G}{2}(1+2\sqrt{2})}$ (B) $\sqrt{\frac{G}{2}(2\sqrt{2}-1)}$

(C) $\sqrt{G(1+2\sqrt{2})}$ (D) $\frac{1}{2}\sqrt{G(1+2\sqrt{2})}$

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Q55. MCQ (Single Correct Answer)

Consider two satellites S_1 and S_2 with periods of revolution 1 hr. and 8 hr. respectively revolving around a planet in circular orbits. The ratio of angular velocity of satellite S_1 to the angular velocity of satellite S_2 is :

Open solution

(A) 1 : 4 (B) 8 : 1 (C) 2 : 1 (D) 1 : 8

Q56. MCQ (Single Correct Answer)

If Y, K and η are the values of Young's modulus, bulk modulus and modulus of rigidity of any material respectively. Choose the correct relation for these parameters.

Open solution

(A) $Y = \frac{9K\eta}{3K-\eta} N/m^2$ (B) $Y = \frac{9K\eta}{2\eta+3K} N/m^2$

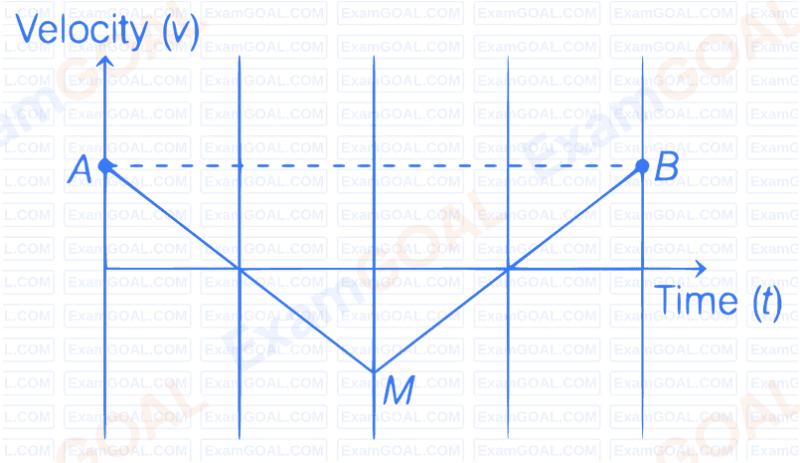
(C) $\eta = \frac{3YK}{9K+Y} N/m^2$ (D) $K = \frac{Y\eta}{9\eta-3Y} N/m^2$

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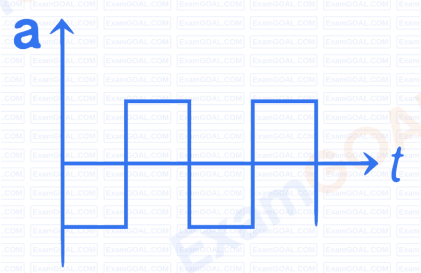
Q57. MCQ (Single Correct Answer)

If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph?

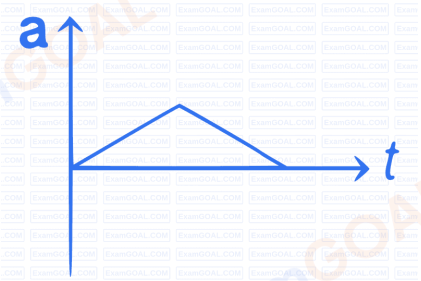
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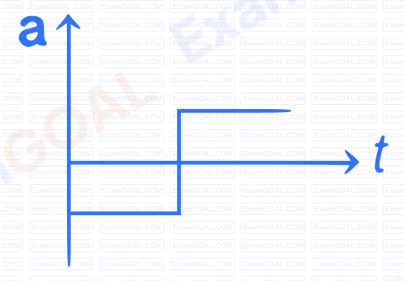
(A)



(B)

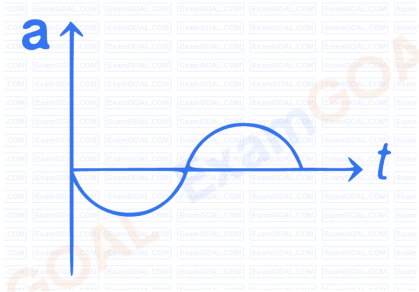


(C)



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(D)

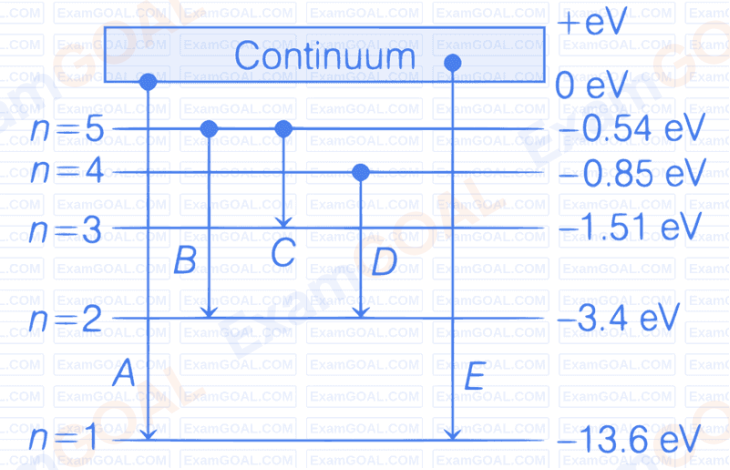


Q58. MCQ (Single Correct Answer)

Open solution

In the given figure, the energy levels of hydrogen atom have been shown along with some transitions marked A, B, C, D and E.

The transitions A, B and C respectively represent :



(A) The series limit of Lyman series, second member of Balmer series and second member of Paschen series.

(B) The ionization potential of hydrogen, second member of Balmer series and third member of Paschen series.

(C) The series limit of Lyman series, third member of Balmer series and second member of Paschen series.

(D) The first member of the Lyman series, third member of Balmer series and second member of Paschen series.

Q59. MCQ (Single Correct Answer)

Open solution

The work done by a gas molecule in an isolated system is

given by, $W = \alpha\beta^2 e^{-\frac{x^2}{\alpha kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. α and β are constants. Then the dimensions of β will be :

(A) $[M^0LT^0]$ (B) $[ML^2T^{-2}]$ (C) $[MLT^{-2}]$ (D) $[M^2LT^2]$

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Q60. MCQ (Single Correct Answer)

Two stars of masses m and 2m at a distance d rotate about their common centre of mass in free space. The period of revolution is :

(A) $\frac{1}{2\pi}\sqrt{\frac{d^3}{3Gm}}$

(B) $2\pi\sqrt{\frac{3Gm}{d^3}}$

(C) $\frac{1}{2\pi}\sqrt{\frac{3Gm}{d^3}}$

(D) $2\pi\sqrt{\frac{d^3}{3Gm}}$

Open solution

Q61. MCQ (Single Correct Answer)

Two equal capacitors are first connected in series and then in parallel. The ratio of the equivalent capacities in the two cases will be :

(A) 4 : 1 (B) 1 : 2 (C) 2 : 1 (D) 1 : 4

Open solution

Q62. MCQ (Single Correct Answer)

Moment of inertia (M. I.) of four bodies, having same mass and radius, are reported as;

I_1 = M.I. of thin circular ring about its diameter,

I_2 = M.I. of circular disc about an axis perpendicular to disc and going through the centre,

I_3 = M.I. of solid cylinder about its axis and

I_4 = M.I. of solid sphere about its diameter.

Then :

(A) $I_1 = I_2 = I_3 > I_4$

(B) $I_1 + I_3 < I_2 + I_4$

(C) $I_1 = I_2 = I_3 < I_4$

(D) $I_1 + I_2 = I_3 + \frac{5}{2} I_4$

Open solution

Q63. Numerical

An unpolarised light beam is incident on the polarizer of a polarization experiment and the intensity of light beam emerging from the analyzer is measured as 100 Lumens. Now, if the analyzer is rotated around the horizontal axis (direction of light) by 30° in clockwise direction, the intensity of emerging light will be _____ Lumens.

Open solution

Q64. Numerical

A ball with a speed of 9 m/s collides with another identical ball at rest. After the collision, the direction of each ball makes an angle of 30° with the original direction. The ratio of velocities of the balls after collision is x : y, where x is _____.

Open solution

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Q65. Numerical

An electromagnetic wave of frequency 5 GHz, is travelling in a medium whose relative electric permittivity and relative magnetic permeability both are 2. Its velocity in this medium is _____ $\times 10^7$ m/s.

Open solution

Q66. Numerical

A hydraulic press can lift 100 kg when a mass 'm' is placed on the smaller piston. It can lift _____ kg when the diameter of the larger piston is increased by 4 times and that of the smaller piston is decreased by 4 times keeping the same mass 'm' on the smaller piston.

Open solution

Q67. Numerical

An inclined plane is bent in such a way that the vertical cross-section is given by $y = \frac{x^2}{4}$ where y is in vertical and x in horizontal direction. If the upper surface of this curved plane is rough with coefficient of friction μ = 0.5, the maximum height in cm at which a stationary block will not slip downward is _____ cm.

Open solution

Q68. Numerical

A common transistor radio set requires 12 V (D.C.) for its operation. The D.C. source is constructed by using a transformer and a rectifier circuit, which are operated at 220 V (A.C.) on standard domestic A.C. supply. The number of turns of secondary coil are 24, then the number of turns of primary are _____.

Open solution

Q69. Numerical

The coefficient of static friction between a wooden block of mass 0.5 kg and a vertical rough wall is 0.2. The magnitude of horizontal force that should be applied on the block to keep it adhere to the wall will be _____ N. [g = 10 ms^{-2}]

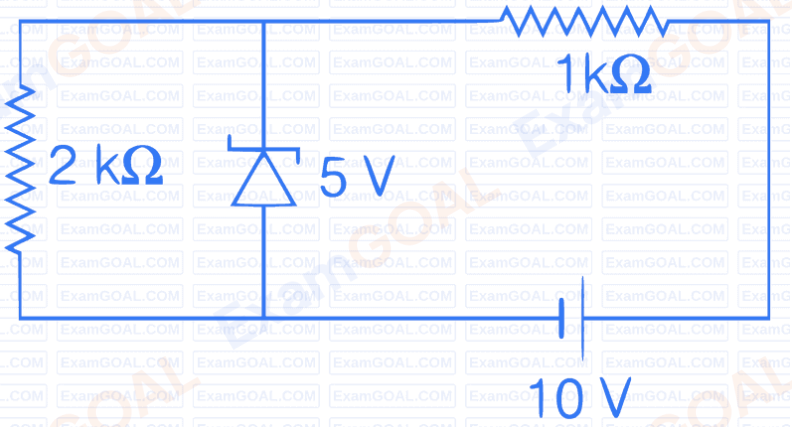
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Q70. Numerical

In connection with the circuit drawn below, the value of current flowing through 2 kΩ resistor is _____ $\times 10^{-4}$ A.

Open solution



Q71. MCQ (Single Correct Answer)

Given below are two statements :

Statement I : Two photons having equal linear momenta have equal wavelengths.

Statement II : If the wavelength of photon is decreased, then the momentum and energy of a photon will also decrease.

In the light of the above statements, choose the correct answer from the options given below.

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Open solution

Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q72. MCQ (Single Correct Answer) · **Chemistry**

The gas released during anaerobic degradation of vegetation may lead to :

(A) Ozone hole

(B) Corrosion of metals

(C) Acid rain

(D) Global warming and cancer

Open solution

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Q73. MCQ (Single Correct Answer) · **Chemistry**

The major components in "Gun Metal" are :

(A) Al, Cu, Mg and Mn

(B) Cu, Ni and Fe

(C) Cu, Sn and Zn

(D) Cu, Zn and Ni

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Match **List I** with **List II**.

List I (Monomer Unit)	**List II (Polymer)**
(a) Caprolactam	(i) Natural rubber
(b) 2-Chloro-1, 3-butadiene	(ii) Buna-N
(c) Isoprene	(iii) Nylon 6
(d) Acrylonitrile	(iv) Neoprene

Choose the correct answer from the options given below:

(A) (a) → (iv), (b) → (iii), (c) → (ii), (d) → (i)

(B) (a) → (iii), (b) → (iv), (c) → (i), (d) → (ii)

(C) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

(D) (a) → (i), (b) → (ii), (c) → (iii), (d) → (iv)

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

(A) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

(B) $I_2 + H_2O_2 + 2OH^- \rightarrow 2I^- + 2H_2O + O_2$

Choose the correct option.

(A) H_2O_2 acts as reducing and oxidising agent respectively in equations (A) and (B).

(B) H_2O_2 acts as oxidizing and reducing agent respectively in equations (A) and (B).

(C) H_2O_2 acts as oxidising agent in equations (A) and (B).

(D) H_2O_2 acts as reducing agent in equations (A) and (B).

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Which of the following ore is concentrated using group 1 cyanide salt ?

(A) Sphalerite **(B)** Malachite **(C)** Siderite **(D)** Calamine

Open solution

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Q77. MCQ (Single Correct Answer) · **Chemistry**

Al_2O_3 was leached with alkali to get X. The solution of X on passing of gas Y, forms Z. X, Y and Z respectively are :

(A) X = $Na[Al(OH)_4]$, Y = CO_2, Z = $Al_2O_3.xH_2O$

(B) X = $Al(OH)_3$, Y = CO_2, Z = Al_2O_3

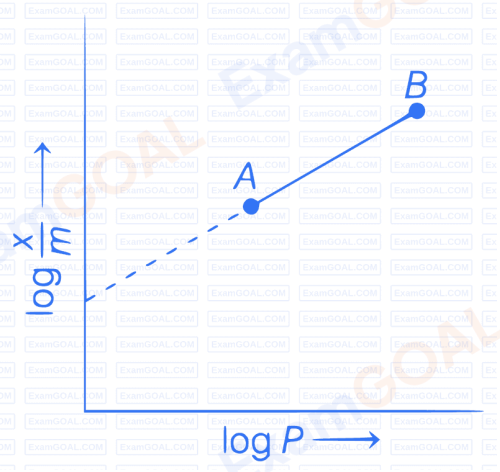
(C) X = $Al(OH)_3$, Y = SO_2, Z = $Al_2O_3.xH_2O$

(D) X = $Na[Al(OH)_4]$, Y = SO_2, Z = Al_2O_3

Open solution

Q78. MCQ (Single Correct Answer) · **Chemistry**

In Freundlich adsorption isotherm, slope of AB line is :



(A) $\frac{1}{n}$ with $\left(\frac{1}{n} = 0 to 1\right)$

(B) n with (n, 0.1 to 0.5)

(C) $\log \frac{1}{n}$ with $(n < 1)$

(D) log n with (n > 1)

Open solution

Q79. Numerical · **Chemistry**

The coordination number of an atom in a body-centered cubic structure is _____.
[Assume that the lattice is made up of atoms.]

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

The equation of the plane passing through the point (1, 2, -3) and perpendicular to the planes

3x + y - 2z = 5 and 2x - 5y - z = 7, is :

(A) 6x - 5y + 2z + 10 =0

(B) 3x - 10y - 2z + 11 = 0

(C) 6x - 5y - 2z - 2 = 0

(D) 11x + y + 17z + 38 = 0

Open solution

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Q81. MCQ (Single Correct Answer) · **Mathematics**

The distance of the point (1, 1, 9) from the point of intersection of the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane x + y + z = 17 is :

(A) $19\sqrt{2}$ (B) $2\sqrt{19}$ (C) 38 (D) $\sqrt{38}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The value of
$-{}^{15}C_1 + 2.{}^{15}C_2 - 3.{}^{15}C_3 + \ldots - 15.{}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \ldots + {}^{14}C_{11}$ is :

(A) $2^{13} - 13$ (B) $2^{16} - 1$ (C) 2^{14} (D) $2^{13} - 14$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Two vertical poles are 150 m apart and the height of one is three times that of the other. If from the middle point of the line joining their feet, an observer finds the angles of elevation of their tops to be complementary, then the height of the shorter pole (in meters) is :

(A) 30 (B) 25 (C) $20\sqrt{3}$ (D) $25\sqrt{3}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

If the tangent to the curve $y = x^3$ at the point $P(t, t^3)$ meets the curve again at Q, then the ordinate of the point which divides PQ internally in the ratio 1 : 2 is :

(A) 0 (B) $2t^3$ (C) $-2t^3$ (D) $-t^3$

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

An ordinary dice is rolled for a certain number of times. If the probability of getting an odd number 2 times is equal to the probability of getting an even number 3 times, then the probability of getting an odd number for odd number of times is :

(A) $\frac{5}{36}$ (B) $\frac{3}{16}$ (C) $\frac{1}{2}$ (D) $\frac{1}{32}$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

The statement among the following that is a tautology is :

(A) $B \rightarrow [A \wedge (A \rightarrow B)]$ (B) $[A \wedge (A \rightarrow B)] \rightarrow B$
(C) $[A \wedge (A \vee B)]$ (D) $[A \vee (A \wedge B)]$

Open solution

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Q87. Numerical · **Mathematics**

Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ be such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$, $\vec{a}.\vec{c} = 7$ and $\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = -\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, then the value of $2\left|\vec{a} + \vec{b} + \vec{c}\right|^2$ is _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

If an emitter current is changed by 4 mA, the collector current changes by 3.5 mA. The value of β will be :

(A) 0.875 (B) 0.5 (C) 3.5 (D) 7

Open solution

Q89. Numerical · **Physics**

An audio signal $v_m = 20 \sin 2\pi(1500t)$ amplitude modulates a carrier $v_c = 80 \sin 2\pi(100{,}000t)$.

The value of percent modulation is _____.

Open solution

Q90. Numerical · **Physics**

A resonance circuit having inductance and resistance 2×10^{-4} H and 6.28Ω respectively oscillates at 10 MHz frequency. The value of quality factor of this resonator is _____. [π = 3.14]

Open solution

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Answer Key

Chemistry

1. **C**	2. **A**	3. **C**	4. **A**	5. **D**
6. **C**	7. **C**	8. **C**	9. **D**	10. **D**
11. **B**	12. **A**	13. **A**	14. **2**	15. **1**
16. **2**	17. **5**	18. **34**	19. **2**	20. **12**
21. **1380**	22. **26**			

Mathematics

23. **B**	24. **A**	25. **C**	26. **A**	27. **C**
28. **C**	29. **D**	30. **D**	31. **C**	32. **C**
33. **D**	34. **A**	35. **B**	36. **17**	37. **9**
38. **10**	39. **1**	40. **5**	41. **3**	42. **3**
43. **540**	44. **6**			

Physics

45. **D**	46. **D**	47. **D**	48. **1**	49. **C**
50. **A**	51. **C**	52. **C**	53. **A**	54. **D**
55. **B**	56. **D**	57. **C**	58. **C**	59. **C**
60. **D**	61. **D**	62. **A**	63. **75**	64. **1**
65. **15**	66. **25600**	67. **25**	68. **440**	69. **25**
70. **25**	71. **D**			

Out of syllabus

72. **D**	73. **C**	74. **B**	75. **D**	76. **A**
77. **A**	78. **A**	79. **8**	80. **D**	81. **D**
82. **D**	83. **D**	84. **C**	85. **C**	86. **B**
87. **75**	88. **D**	89. **25**	90. **2000**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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